Livia Chandra

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EDUCATION

University of California – San Diego **Sep 2025 – Mar 2027**
Master of Science, Electrical and Computer Engineering (GPA: 3.7) *La Jolla, CA*
- **Coursework:** Systems for LLMs and AI Agents (in progress), Parallel Computing, Advanced Computer Architecture, Recommender Systems

University of Wisconsin – Madison **Sep 2022 – May 2025**
Bachelor of Science, Computer Science and Data Science (GPA: 3.9) *Madison, WI*
- **Coursework:** Database Management Systems, Big Data Systems, Software Engineering, Algorithms, Machine Organization

SKILLS

- **Languages:** Java, C++, Python, TypeScript, JavaScript, C#, C, SQL, CUDA, Bash, R, Verilog, Go
- **Web/Backend:** React, REST APIs, GraphQL, gRPC, Protocol Buffers (protobuf), Model Context Protocol (MCP) servers, Microservices, Spring Boot, NestJS, Flask, TypeORM, MySQL, PostgreSQL
- **Tools:** Git, Docker, Linux, CMake, Nsight, Thrust, Valgrind, AWS, llama.cpp, JUnit, Claude Code
- **Machine Learning:** PyTorch, scikit-learn
- **Concepts:** Data Structures, Software Design, Data Modeling, Database Design, API Integration, RAG, LLM Application Development, AI Evaluation, Distributed Systems, Operating Systems, Multithreading, Object-Oriented Programming and Design, Design Patterns, Agile Development
- **Soft Skills:** Technical Communication, Cross-functional Collaboration, Problem Solving

EXPERIENCE

UC San Diego Turakhia Lab **May 2026 – Present**
Graduate Student Researcher | C++, CMake, Linux, Intel TBB *La Jolla, CA*
- Building a scalable C++ metagenomic classifier for larger and longer reference sequences, adopting Kraken 2's compact hash table and sampling to match its index size (100 GB) and query runtime.
- Halved query peak memory vs. Kraken 2 (50 vs. 106 GB), which loads its whole index into RAM, by serving a zero-copy memory-mapped index with hierarchical sharding and Bloom-filter routing.
- Raised precision over Kraken 2 (0.928 vs. 0.880) by implementing the Fitch algorithm to encode state changes compactly per key, recovering information Kraken 2 discards.
- Cut full build time 7.7× (1.5 vs. 11.5 h) with a streaming Intel TBB pipeline that parallelizes reading, merging, and Fitch encoding, using sorted runs and a range-partitioned k-way merge instead of Kraken 2's lock-contended shared hash table.

UW–Madison Speech Processing and Auditory Neuroscience Laboratory **Sep 2024 – May 2025**
Undergraduate Student Researcher | Python, PyQt5, PyInstaller, GitHub Actions *Madison, WI*
- Built a cross-platform PyQt5 app that ran Mandarin tone perception and production experiments for 20 participants, with seaborn dashboards tracking performance within and across sessions.
- Automated Windows and macOS releases with a GitHub Actions and PyInstaller CI pipeline that ships a standalone build on every push, letting 10 non-programmer researchers run the app without any setup.

GGCG **Aug 2024 – Sep 2024**
Software Engineer Intern | NestJS, TypeScript, GraphQL, TypeORM, PostgreSQL *Remote*
- Consolidated duplicated address logic in a membership services portal into a shared GraphQL address module with cascading country, state, and city selection, serving case-insensitive search over 19K regions in under 50 ms.
- Extended the outlet GraphQL API with geolocation, background-asset fields, and partial-update mutations, powering the portal's outlet map and the launch of a new service reservation feature.

PROJECTS

Research Agent | *Java, Spring Boot, PostgreSQL, pgvector, React, JUnit* Code **Sep 2026 – Present**
- Built a RAG research assistant as 5 Spring Boot microservices, serving pgvector semantic search over 500 papers at under 500 ms p95 (vector lookup under 60 ms) and summarizing related work through a pluggable LLM API layer.
- Designed the PostgreSQL schema and REST APIs that persist users' research decisions, powering a React dashboard that tracks progress and flags overlooked caveats.
- Covered the backend services with 100 JUnit tests (80% coverage).

Equipment Rental Company Website Rebuild | *Next.js, TypeScript, PostgreSQL, Go* **Nov 2026 – Present**
- Rebuilding a Malaysian equipment rental company's 2002-era PHP site as a Next.js and TypeScript catalog of 8 equipment lines, targeting a Largest Contentful Paint (LCP) under 2 s on 4G and a Lighthouse score of 90+.
- Designing the PostgreSQL schema and Zod-validated catalog API shared by every page and service, targeting responses under 200 ms at p95.
- Building an owner dashboard with two-factor authentication (2FA) for updating prices and availability, where each save triggers on-demand revalidation of only the affected product and catalog pages to publish edits within 5 s without a full rebuild.
- Building a Go gateway for a customer-facing LLM chatbot that recommends equipment from the catalog, enforcing per-IP rate limits and checking every suggested product against the database, targeting a first reply under 3 s.

Multimodal Inference Routing on Heterogeneous Edge Machines | *Python, llama.cpp, AWS EC2* Code **Oct 2026 – Dec 2026**
- Benchmarked llama.cpp inference of Qwen2.5-Omni-3B on Intel (AMX), AMD, and Graviton4 EC2 instances, finding that AMX speeds up prefill by 42% for text, 29% for audio, and 24% for images.
- Designed a workload-aware router that estimates request cost from modality and input size, based on profiling showing input encoding dominates latency (audio: 38 of 45.2 s TTFT), cutting p95 latency by 58% at 60% load and 72% at 90% load vs. round-robin.

Parallelization of a Phylogenetic Tree Method | *CUDA, C++* Code **Mar 2026**
- Parallelized Neighbor-Joining in CUDA by block-level parallel reduction for the min-pair search and O(N) incremental distance updates, achieving a 20× speedup over QuickTree on a 10,000-taxon tree with comparable accuracy.